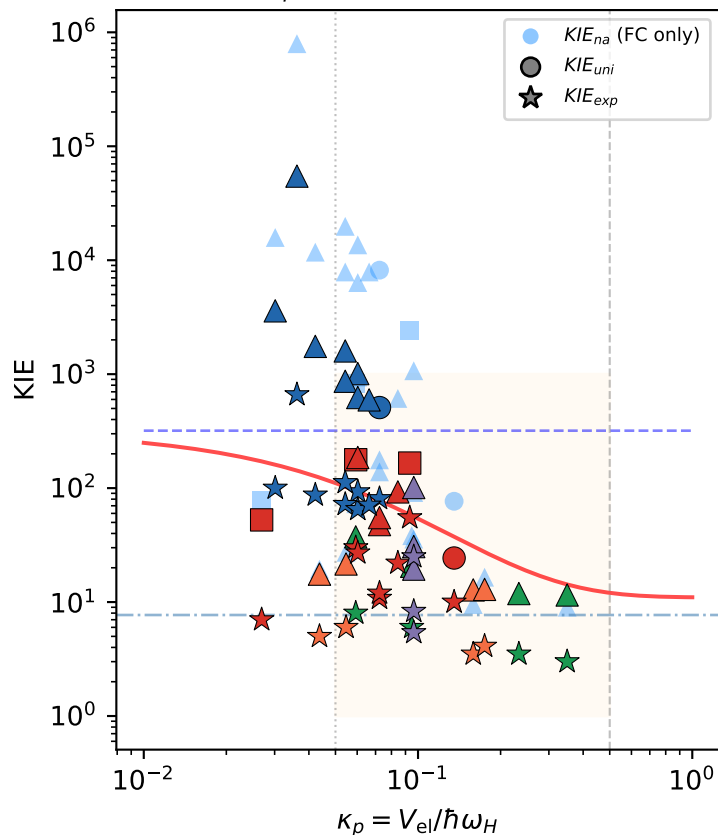
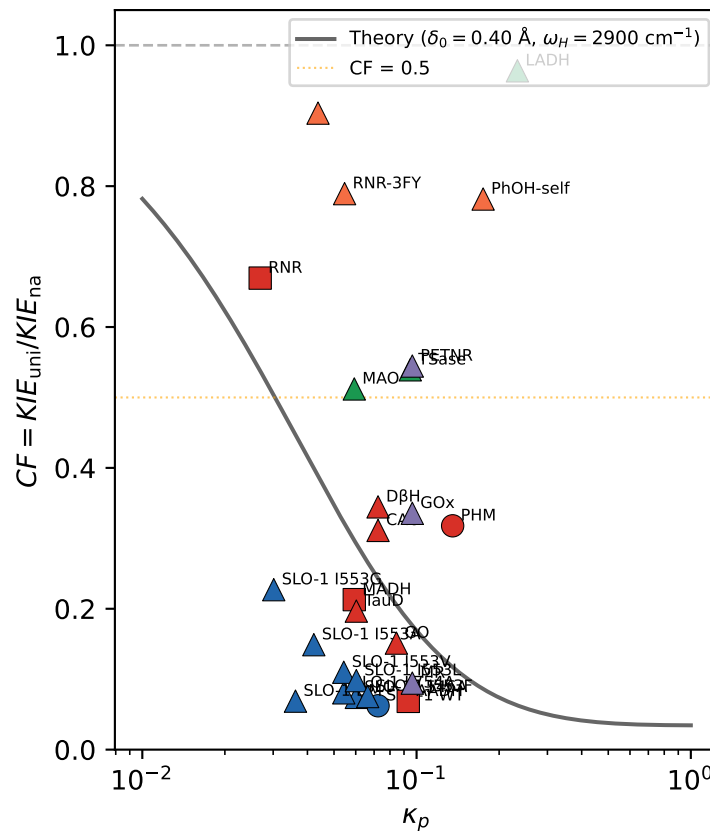


Unified adiabatic-nonadiabatic PCET KIE: 28-system benchmark

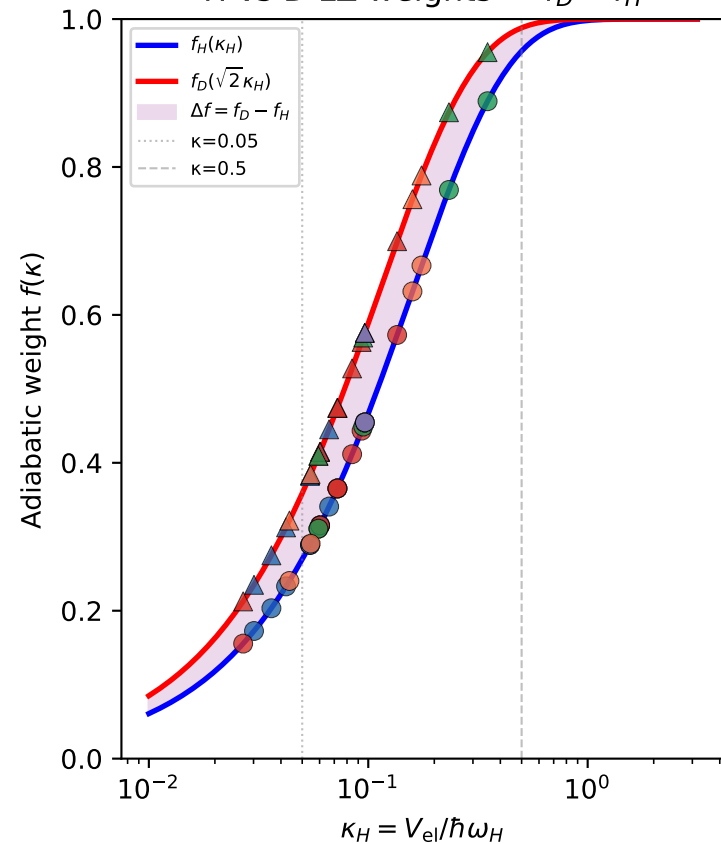
KIE vs κ_p — NA, unified, and exp



Adiabatic correction to KIE



H vs D LZ weights — $f_D > f_H$



Legend for enzyme families:

- SLO-1 family (Blue)
- HAT (Red)
- Hydride (Green)
- Proton/O-H (Orange)
- Flavin (Purple)